

1.) Consider the training example shown in Table 1 for a binary classification problem

Table 1. Data set $\mathcal{D}$			
Movie ID	Format	Movie Category	Class
1	DVD	Entertainment	C0
2	DVD	Comedy	C0
3	DVD	Documentaries	C0
4	DVD	Comedy	C0
5	DVD	Comedy	C0
6	DVD	Comedy	C0
7	Online	Comedy	C0
8	Online	Comedy	C0
9	Online	Comedy	C0
10	Online	Documentaries	C0
11	DVD	Comedy	C1
12	DVD	Entertainment	C1
13	Online	Entertainment	C1
14	Online	Documentaries	C1
15	Online	Documentaries	C1
16	Online	Documentaries	C1
17	Online	Documentaries	C1
18	Online	Entertainment	C1
19	Online	Documentaries	C1
20	Online	Documentaries	C1

(a) Compute the Entropy for the overall collection of training examples

Solution (a)

$$\text{Entropy} = - \sum_{i=0}^{c-1} P_i(t) \log_2 P_i(t)$$

From the given data set in Table 1

Total instances = 20

$$C_0 = 10 \text{ instances} \quad \therefore P(C_0) = \frac{10}{20} = \frac{1}{2}$$

$$C_1 = 10 \text{ instances} \quad \therefore P(C_1) = \frac{10}{20} = \frac{1}{2}$$

$$\text{Entropy} = - [P(C_0) \log_2 P(C_0) + P(C_1) \log_2 P(C_1)]$$

$$= - \left[\left(\frac{1}{2}\right) \log_2 \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right) \log_2 \left(\frac{1}{2}\right) \right]$$

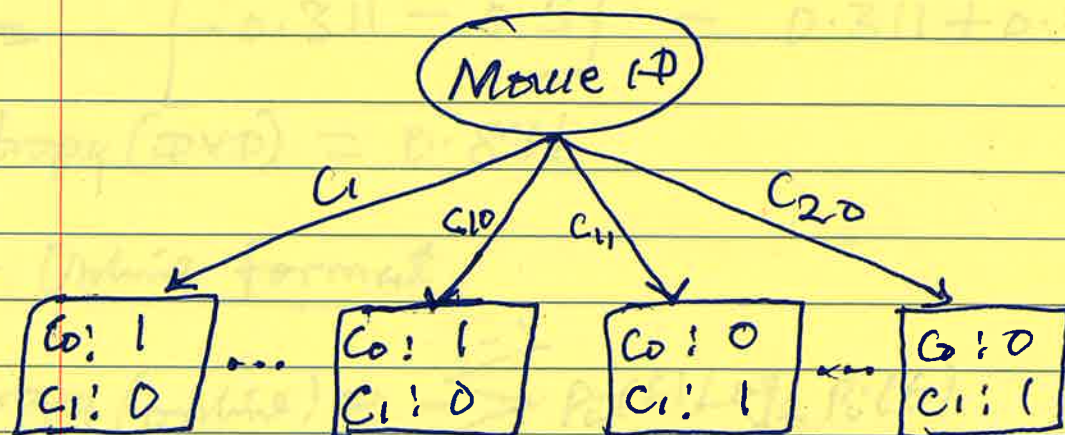
$$= - \left[\left(\frac{1}{2}\right) (-1) + \left(\frac{1}{2}\right) (-1) \right]$$

$$= - \left[-\frac{1}{2} - \frac{1}{2} \right] = \frac{1}{2} + \frac{1}{2} = \frac{2}{2} = 1$$

$$\therefore \text{Entropy} = 1$$

(b) Compute the Entropy for the Movie ID attribute

Solution (b)



$$\text{Entropy (Movie ID)} = - \sum_{i=0}^{c-1} P_i(t) \log_2 P_i(t)$$

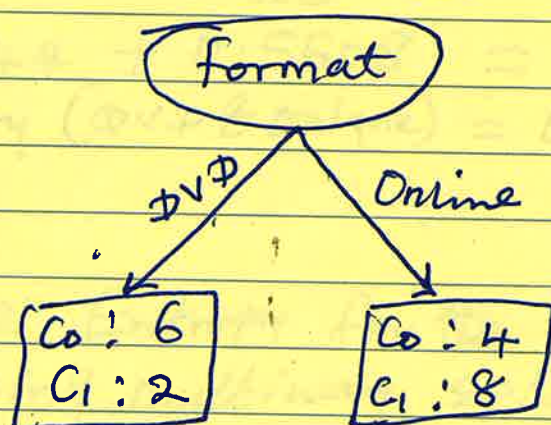
$$\text{Entropy (Movie ID)} = - \left[\left(\frac{0}{1} \right) \log_2 \left(\frac{0}{1} \right) + \left(\frac{1}{1} \right) \log_2 \left(\frac{1}{1} \right) \right]$$

$$= - \left[0 \log_2(0) + 1 \log_2(1) \right] = - [0 + 0] = 0$$

$$\text{Entropy (Movie ID)} = 0.$$

(C) Compute the Entropy for the Format attribute.

Solution (C)



For DVD format:

$$\text{Entropy (DVD)} = - \sum_{i=0}^{c-1} p_i(t) \log_2 p_i(t)$$

$$= - \left[\left(\frac{6}{8} \right) \log_2 \left(\frac{6}{8} \right) + \left(\frac{2}{8} \right) \log_2 \left(\frac{2}{8} \right) \right]$$

$$= - \left[\left(\frac{6}{8} \right) (-0.415) + \left(\frac{2}{8} \right) (-2) \right]$$

$$= - [-0.311 - 0.5] = 0.311 + 0.5 = 0.811$$

$$\text{Entropy (DVD)} = 0.811$$

For Online format

$$\text{Entropy (online)} = - \sum_{i=0}^{c-1} p_i(t) \log_2 p_i(t)$$

$$= - \left[\left(\frac{4}{12} \right) \log_2 \left(\frac{4}{12} \right) + \left(\frac{8}{12} \right) \log_2 \left(\frac{8}{12} \right) \right]$$

$$= - \left[\left(\frac{4}{12} \right) (-1.584) + \left(\frac{8}{12} \right) (-0.585) \right]$$

$$= - [-0.528 - 0.39] = 0.528 + 0.39$$

$$= 0.918$$

∴ Entropy (online) = 0.918.

Weighted Entropy of DVD and Online

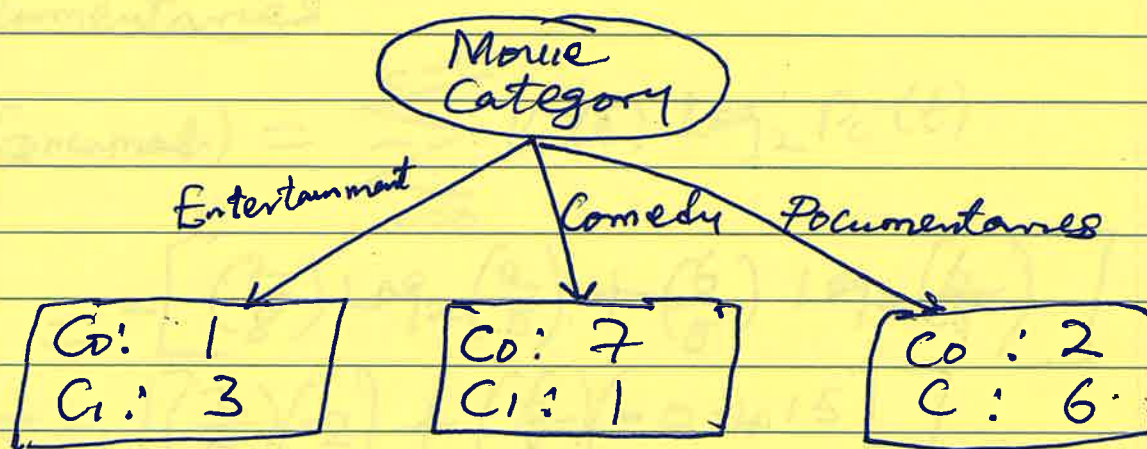
$$= \frac{8}{20} \times 0.811 + \frac{12}{20} \times 0.918$$

$$= 0.3244 + 0.5508 = 0.8752$$

∴ weighted Entropy (DVD & online) = 0.8752

(d) Compute the Entropy for the Movie Category attribute using multiway split.

Solution (d)



For Entertainment:

$$\text{Entropy (Ent.)} = - \sum_{c=0}^{c-1} P_i(c) \log_2 P_i(c)$$

$$= - \left[\left(\frac{1}{4} \right) \log_2 \left(\frac{1}{4} \right) + \left(\frac{3}{4} \right) \log_2 \left(\frac{3}{4} \right) \right]$$

$$= - \left[\left(\frac{1}{4} \right) (-2) + \left(\frac{3}{4} \right) (-0.415) \right]$$

$$= - \left[-\frac{2}{4} - 0.311 \right] = 0.5 + 0.311 =$$

$$\text{Entropy (Ent)} = 0.811$$

For Comedy

$$\text{Entropy (Comedy)} = - \sum_{i=0}^{c-1} P_i(t) \log_2 P_i(t)$$

$$= - \left[\left(\frac{7}{8}\right) \log_2 \left(\frac{7}{8}\right) + \left(\frac{1}{8}\right) \log_2 \left(\frac{1}{8}\right) \right]$$

$$= - \left[\left(\frac{7}{8}\right) (-0.193) + \left(\frac{1}{8}\right) (-3) \right]$$

$$= - \left[-0.1689 - 0.375 \right] = 0.1689 + 0.375$$

$$\text{Entropy (Comedy)} = 0.5439 \approx \underline{0.544}$$

For Documentaries

$$\text{Entropy (Document.)} = - \sum_{i=0}^{c-1} P_i(t) \log_2 P_i(t)$$

$$= - \left[\left(\frac{2}{8}\right) \log_2 \left(\frac{2}{8}\right) + \left(\frac{6}{8}\right) \log_2 \left(\frac{6}{8}\right) \right]$$

$$= - \left[\left(\frac{2}{8}\right) (-2) + \left(\frac{6}{8}\right) (-0.415) \right]$$

$$= - \left[-\frac{4}{8} - 0.31125 \right] =$$

$$= - \left[-0.5 - 0.31125 \right] = 0.5 + 0.31125$$

$$\text{Entropy (Doc)} = 0.81125 \approx 0.811$$

The Weighted Entropy for Movie Category.

$$= \left(\frac{4}{20}\right) \times 0.811 + \left(\frac{8}{20}\right) \times 0.544 + \left(\frac{8}{20}\right) \times 0.811$$

$$= 0.1622 + 0.2176 + 0.3244$$

$$= \underline{\underline{0.7042}}$$

(e) Which of the three attributes has the lowest Entropy?

Solution (e)

The Movie ID has the lowest Entropy which is 0.
 $\therefore \text{Entropy}(\text{Movie ID}) = 0.$

(f) Which of the three attributes will you use for splitting at the root node? Briefly explain your choice.

Solution (f)

To determine which of the three attributes for splitting at the root node, I have to first determine their individual gain information.

$$\text{Gain} = P - M$$

$$\text{Gain}(\text{Movie ID}) = 1 - 0 = 1$$

$$\text{Gain}(\text{Format}) = 1 - 0.8752 = 0.125$$

$$\text{Gain}(\text{Movie Category}) = 1 - 0.7042 = 0.296$$

Based on the calculated information gain values, I would choose Movie Category as the attribute to split on at the root node because it has the higher information gain (0.296) after Movie ID. Although Movie ID has the maximum possible information gain (1), it likely leads to overfitting, since that attribute perfectly partitions the training data.

Movie Category provides a good amount of information gain while likely avoiding overfitting as much as Movie ID. The Format attribute provides relatively little information gain for partitioning the data.

[No2]

Consider the decision tree shown in Figure 1, and the corresponding training and test sets in Table 2 and 3 respectively.

- Estimate the generalization error rate of the tree, to account for model complexity, use a penalty value of 2 to each leaf node.
- Compute the error rate of the tree on the test set shown in Table 3
- Comment on the behavior of training and test set errors with respect to model complexity. Comment on the utility of incorporating model complexity in building a predictive model

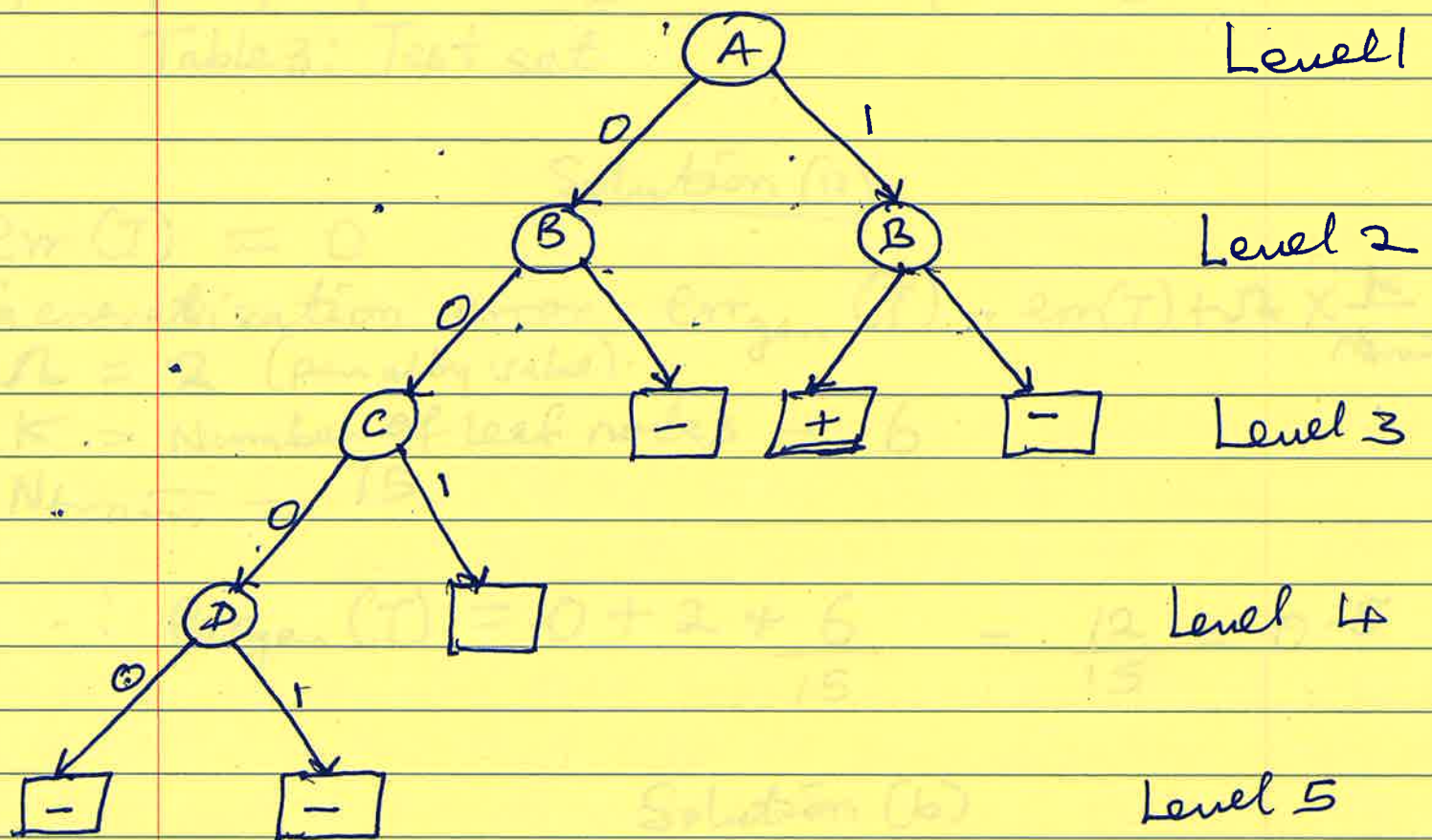


Figure 1: Decision tree

A	B	C	D	Number of + instances	Number of - instances
0	0	0	0	4	0
0	0	0	1	0	1
0	0	1	0	0	1
0	1	0	1	0	1
1	0	1	0	3	0
1	1	0	1	0	5

Table 2: Training set

A	B	C	D	Number of + instances	Number of - instances
0	0	0	1	4	0
0	0	1	1	3	0
0	1	0	0	0	1
0	1	1	0	0	2
1	0	0	0	2	0
1	0	0	1	3	0

Table 3: Test set

Solution (a).

$$\text{err}(T) = 0$$

$$\text{Generalization error } \text{err}_{\text{gen}}(T) = \text{err}(T) + \lambda \times \frac{K}{N_{\text{train}}}$$

$\lambda = 2$ (penalty value).

$K = \text{Number of leaf nodes} = 6$

$N_{\text{train}} = 15$

$$\therefore \text{err}_{\text{gen}}(T) = 0 + 2 \times \frac{6}{15} = \frac{12}{15} = 0.8$$

Solution (b)

$$\text{Test Error} = \frac{4 + 3 + 0 + 0 + 0 + 0}{15} = \frac{7}{15} = 0.47$$

$$\text{Test Error} = 0.47$$

Solution (C) i

* Observed that, as the tree is pruned more and more, the generalization error on the training set increases, but the error on the test set decrease to 0 when level 3 is pruned. This illustrates the overfitting problem at level 3 and 4 of the tree. Pruning further from level 3 to level 2 leads to an increase in both training and test error, since the model becomes too simple and suffers from underfitting.

The answers to my comment are provided in the table below for better understanding.

Level Pruned	Error _{train}	Error _{test}
No pruning	0	$7/15 = 0.47$
4	$1/15 = 0.066$	$3/15 = 0.2$
3	$2/15 = 0.133$	$0/15 = 0$
2	$6/15 = 0.4$	$8/15 = 0.53$

Solution (C) ii

* Comment on the utility of incorporating model complexity in building a predictive model.

The lowest error on the test set is obtained for the pruned version of the tree with 3 levels.

$$\therefore \text{Error at level 3} = (2 + 4 \times 2) / 15 = 2/3 = 0.666$$

The original decision tree suffers from overfitting, which is reflected in the fact that its generalization error on the test set is significantly higher than that on the training set. Trees with smaller sizes are less prone to overfit the data. In fact, compared to the original tree, the pruned tree has a lower error rate. A simpler (pruned) tree can be chosen despite having a larger error rate on the training set because the

error rate in corporate model complexity.

[No 3]

You're trying to determine whether Andrew finds a particular type of food appealing based on the food's temperature, taste, and size. Answer the questions regarding the dataset shown below.

Appealing	Temperature	Taste	Size
No	Hot	Salty	Small
No	Cold	Sweet	Large
No	Cold	Sweet	Large
Yes	Cold	Sour	Small
Yes	H	Sour	Small
No	H	Salty	Large
Yes	H	Sour	Large
Yes	Cold	Sweet	Small
Yes	Cold	Sweet	Small
No	H	Salty	Large

Q What is the initial entropy of Appealing?
Solution (a)

$$\text{Entropy} = - \sum_{i=0}^{c-1} P_i(t) \log_2 P_i(t)$$

Let $c_0 = \text{No}$, $c_1 = \text{Yes}$.

• No = 5, Yes = 5. Total instance = 10.

$$\text{Entropy} = - \left[\left(\frac{5}{10} \right) \log_2 \left(\frac{5}{10} \right) + \left(\frac{5}{10} \right) \log_2 \left(\frac{5}{10} \right) \right]$$

$$\text{Entropy} = - \left[\left(\frac{5}{10} \right) (-1) + \left(\frac{5}{10} \right) (-1) \right]$$

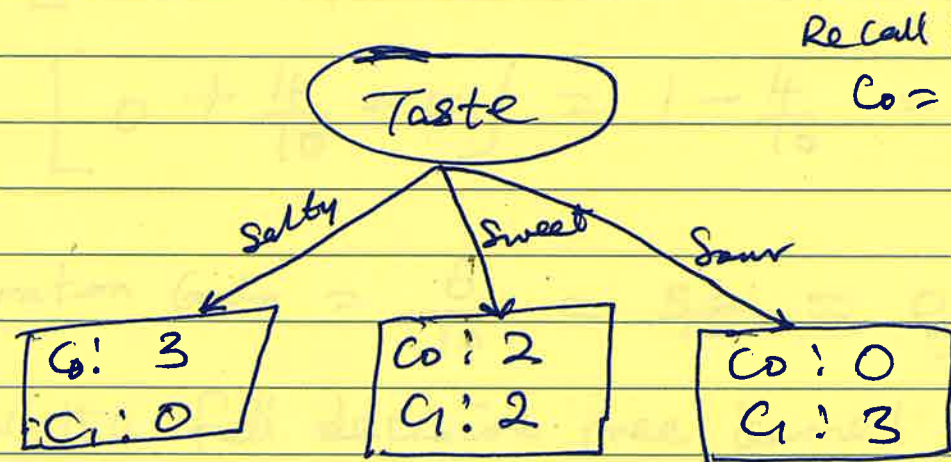
$$\text{Entropy} = - \left[-\frac{5}{10} - \frac{5}{10} \right] = \frac{5}{10} + \frac{5}{10} = \frac{10}{10}$$

$$\text{Entropy} = \underline{\underline{1}}$$

over 1000 or over 1000 ways of computing it.

(b) Assume that Taste is chosen for the root of the decision tree. What is the information gain associated with this attribute?

Entropy (parent) = 1 Solution (b)
 Entropy (children) calculated.



$$\text{Entropy (Salty)} = - \sum_{c=0}^{c=1} P_S(c) \log_2 P_S(c)$$

$$= - \left[\left(\frac{3}{3}\right) \log_2 \left(\frac{3}{3}\right) + \left(\frac{0}{3}\right) \log_2 \left(\frac{0}{3}\right) \right]$$

$$= - \left[\left(\frac{3}{3}\right)(0) + 0 \right] = [0] = 0$$

$$\text{Entropy (Sweet)} = - \left[\left(\frac{2}{4}\right) \log_2 \left(\frac{2}{4}\right) + \left(\frac{2}{4}\right) \log_2 \left(\frac{2}{4}\right) \right]$$

$$= - \left[\left(\frac{2}{4}\right)(-1) + \left(\frac{2}{4}\right)(-1) \right]$$

$$= - \left[-\frac{2}{4} - \frac{2}{4} \right] = \frac{2}{4} + \frac{2}{4} = \frac{4}{4} = 1$$

$$\text{Entropy (Sour)} = - \left[\left(\frac{0}{3}\right) \log_2 \left(\frac{0}{3}\right) + \left(\frac{3}{3}\right) \log_2 \left(\frac{3}{3}\right) \right]$$

$$= - [0 + 1(0)] = 0$$

Information Gain

$$= \text{Entropy}(\text{parent}) - \sum_{c=1}^k \frac{n_c}{n} \cdot \text{Entropy}(c)$$

$$= 1 - \left[\left(\frac{3}{10} \right) \cdot \text{Entropy}(\text{salty}) + \left(\frac{4}{10} \right) \cdot \text{Entropy}(\text{sweet}) + \left(\frac{3}{10} \right) \cdot \text{Entropy}(\text{sour}) \right]$$

plugging salty, sweet and sour Entropy values we have.

$$= 1 - \left[\left(\frac{3}{10} \right) \times 0 + \frac{4}{10} \times 1 + \frac{3}{10} \times 0 \right]$$

$$= 1 - \left[0 + \frac{4}{10} + 0 \right] = 1 - \frac{4}{10} = \frac{10-4}{10}$$

$$\therefore \text{Information Gain} = \frac{6}{10} = \frac{3}{5} = \underline{\underline{0.6}}$$

(C) Draw the full decision tree learned for this data (without any pruning)

